

Android 2.2 unveiled

Google has unveiled Android 2.2 with full flash support, even as daily sales for Android devices cross the 1,00,000 mark

Corsair's new SSD

Corsair's Force family of SSDs now has a maximum capacity of 240 GB

When a 3D image is displayed on screen, each lens will display a slightly different perspective; that is each image is slightly offset from the other. In this regard these glasses work similarly to passive 3D glasses. But this is where the similarity ends – both images aren't displayed at the same time on the screen. The display turns each image on and off at very high speeds, therefore with the naked eye it would appear as if there are two slightly offset, identical images on the screen. The LCD glasses alternate between being transparent and opaque in sync with the alternating image on the screen. When the image meant for the right eye appears on screen, the left lens on the LCD glasses will switch off blacking out the view to

the screen for the left eye and vice versa. This blacking out of the LCD glasses happens very fast, fast enough to fool your mind into believing its not happening, since your eyes cannot spot up the difference. This process is synchronised perfectly with the screen, therefore each eye sees only one set of

the two offset images. Obviously this sort of fast image refreshing requires a TV with a higher refresh rate than the de facto 60-Hertz. 3D TVs have a refresh rate of at least 120-Hertz to be able to utilise this technology. But that is not the only prerequisite. So HDTVs today that support up to 200 Hertz refresh rates may not be 3D ready – the

high refresh rate is merely being advertised as a means to control the action in really fast scenes and to allow for less distortion in the image when using either NTSC TV (30 fps) or cinema/movies (24 fps) standards.

In India, active 3D glasses is where all the action is promising to hot up – a lot of

brands are promising such 3D HDTVs, but there are very few models available. Samsung already has an entire 3D range and are to be commended for making it to the market first. Their 7000/8000 series LED-backlit HDTVs and certain other LCD and plasma displays support 3D. They've also announced plans to create an ecosystem of 3D Blu-ray players, obviously to speed up market adoption of 3D products. Major players in India like LG and Sony 3D displays very soon. In fact LG has such products in the US market, but the cost associated with buying such displays along with the cost of accompaniments like 3D glasses, a 3D-ready Blu-ray player and such will surely be deterrents in a price sensitive market. LG has an upcoming Infinia LED Series, that feature full 3D support. All of these displays from Samsung, and upcoming ones from LG and Sony will need active 3D glasses.

One very interesting option being pursued by Mitsubishi is the use of a DLP rear projection-based setup for a 3D HDTV. One of the benefits of this DLP-projection based technology that has been licensed from Texas Instruments is that it is cheaper than current LCD panel technology. It therefore allows larger panels, to the tune of 65-inches to be available at the same price as 47-inch LCD panels. However, although the technology going into the display is different, the same 3D glasses are required for viewing 3D content properly. They also have a 65-inch Laser TV – where laser light is used to project 3D images on the screen.

Auto-stereoscopic displays: the way it's meant to be

We have used NVIDIA's 3D Stereo Vision for gaming, so we've sampled what 3D has to offer and it is most impressive, but not a technology that has fully arrived, meaning that in it's current state, we feel there's a lot of work to be

done. For one, who likes to wear nerdy glasses? 3D glasses can be restrictive, not to mention there are some inhibitions about bolting on a pair of LCD screens to your eyes and the long term effects on a persons general ocular-health. Also, such glasses are expensive, the NVIDIA Stereo Vision costs around Rs. 11,000. What happens if a family of five needs to watch a 3D movie? Also such displays require tremendous amounts of video bandwidth to be fed to them, since they are drawing two identical images on screen. While watching 1080p content in 3D, for example, it is necessary for the display to support a resolution of at least 2160p. The higher the resolution, the clearer the image and the more realistic the 3D view.

Such displays have been around for a while, but there is nothing regularly available in the consumer space because of the number of conditions required for getting a proper 3D view without glasses. For example, the audience needs to be in a fixed location or area and any movement will spoil the 3D effect. Another problem is bandwidth – some researchers estimate the bandwidth required could be as high as 200 gigabits per second – this is not possible today. The resolution required would also be very high, something not possible on today's LCD and plasma displays.

